WASTEWISE SCAN · RISK REGISTER AND HAZARD ANALYSIS

WasteWise Scan risk register and hazard analysis

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	First risk register, preliminary hazard analysis and first-pass design FMEA (kit 1.8 upgrade, stage P3) for option A

WasteWise Scan risk register and hazard analysis

The top project risks are that eight bands cannot tell PE from PP to the 95 % target, and that the SWIR LEDs and InGaAs photodiode cost far more than the BOM's indicative prices. The top hazards are the lithium-ion cell (heat, drop, charging in sun), invisible LED light if a driver sticks on, and misuse of a resin result to judge whether waste is safe to handle. The highest design FMEA items are a wrong but confident resin call and a leaking baffle tube. Method: STANDARDS section 23. Likelihood, impact, severity and occurrence are scored 1 (low) to 5 (high); detection 1 (certain to be caught) to 5 (not caught).

> **Safety:** The scanner holds a lithium-ion cell of about 11 Wh and drives invisible infrared LEDs. It identifies resin only and must never be used to judge whether waste is safe to handle. See WSC-PRC-001, Safety, and the safety stops in WSC-BLD-001.

Project risk register

Table 1. Project risks.

#	Risk	Cause	Effect	Likelihood	Impact	Owner (role)	Response
P1	PE and PP cannot be told apart to 95 %	1,650 nm band clipped by the detector cutoff; real item scatter above 1 %	PR-001 not met; users lose trust	3	4	Design lead	Labelled item set first at TRL 4; ninth band with extended InGaAs ready as an option (WSC-DEC-001 open 1); option B as the fallback principle (WSC-OPT-001)
P2	Critical optics cost more than the BOM shows	SWIR LED prices indicative; a named 1 mm InGaAs photodiode lists at EUR 104 against USD 30	Cost far over the USD 150 target	3	3	Procurement	Quotes from two suppliers; LED set from one maker; six-band saving if items allow (WSC-DEC-001, Value engineering)
P3	No co-design partner engaged	Outreach to SWaCH not yet made	Requirements unconfirmed; no item set or price table	3	4	Project lead (Amish)	Approach SWaCH jointly with WasteWise-ml (decided 2026-10-02)
P4	WasteWise-ml does not adopt the shared record	Separate project timelines	PR-013 stays at risk	2	2	Project lead	Joint interface note (WSC-DDR-002)
P5	Radio or cell approvals block a pilot in India	WPC ETA for the board's module; BIS registration of cells	Delay to any field use	2	3	Procurement (partner)	Buy approved modules and registered cells (WSC-REG-001 R3, R5)
P6	Display unreadable in direct sun	Hood cannot shade every sun angle	PR-007 at risk	3	2	Design lead	Colour and icon cues; user guidance on holding angle; design review with users
P7	Mass over 250 g	Parts added for construction	PR-006 not met by 9 g	4	1	Design lead	Weigh at TRL 4; trim frame and head block if it matters to users

Preliminary hazard analysis

Table 2. Hazards.

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
H1	Lithium-ion cell fire or venting	Charging in sun or above 45 C; crushed or dented cell; cell loose in a drop	Burns, smoke	5	2	Protected cell from a named maker (BIS registered); board charger only; no charging above 45 C or in sun; cradle and strap hold the cell against 869 N peak (PR-015, WSC-CAL-001 J4, fix F3); safety stops S1 and S4 in WSC-BLD-001	Low, if the charging rule is followed
H2	Invisible infrared light to the eye	LED driver stuck on; looking into the window	Corneal or retinal heating over long exposure	3	1	Exempt group in normal use (3.9 W/m2 at 200 mm); hardware on-time limit that firmware cannot override; label "do not look into the window"	Very low
H3	Resin result used to judge hazardous waste	User trusts the scanner beyond its purpose	Cuts, needle-stick, chemical exposure	4	2	Start-screen warning; "unknown" rather than a guess; out-of-scope statement in all user material	Low
H4	Hand near sharp items when backing clear items with the cap	Backing step in sun	Cuts	3	2	Scan sharp or broken items in shade instead; gloves	Low
H5	Burns from the soldering iron when setting inserts (build)	230 C iron	Burns	2	2	Stand, clamped work, build plan safety notes	Low
H6	Glass fragments when core-drilling the window (build, fix F1)	Diamond core drill on borosilicate	Cuts, eye injury	3	2	Wet drilling at low speed, disc clamped on scrap wood, safety glasses, deburr the edge	Low
H7	Wrong sorting decisions	Confident wrong result	Loss of income, contaminated bales	2	3	Confidence threshold set for few false results (PR-003); white reference check; "unknown" for black items	Medium until item data exists

Design FMEA (first pass, P3)

Table 3. Failure modes. RPN = severity x occurrence x detection.

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F1	Classifier and band set	Name the resin	Confident wrong call between PE and PP	Wrong pile, lost value	Clipped 1,650 nm band; item scatter	5	3	4	60	Labelled item set; confidence threshold tuned for few false results; ninth band option
F2	Baffle tube (fix F1)	Keep LED light from reaching the detector inside the glass	Tube passes near-infrared light	Offset changes with dirt; wrong ratios	Dye-coloured black filament	4	2	3	24	Carbon-black filament; 1 mm sample check (WSC-BLD-001 3.3); open-air reading still taken
F3	Detector window glue joint	Hold the 5.8 mm window flush in the tube	Window loose or glue on the optical face	Light leak or signal loss	Poor bead; drop	3	2	2	12	Edge bead only; check under magnification; white reference shows signal loss
F4	LED driver	Pulse each LED for about 14 ms	Stuck on	Eye exposure; battery drain	Firmware fault or failed switch	4	2	3	24	Hardware on-time limit; current monitor
F5	Cell strap and inserts	Hold the cell in a 1.2 m drop	Strap cracks along layers or insert pulls out	Cell loose; contacts open	Printed PETG strength; margin 1.45 at the peak (fix F3) on an assumed crush	4	2	3	24	Print with layers along the bar; drop test at TRL 4; thicker strap if it fails
F6	Shell gasket and button seal	Keep dust and rain out (IP54)	Water ingress	Corrosion, short	Gasket over- or under-squeezed	3	3	3	27	Bosses set the squeeze at 0.6 mm; spray test at TRL 4
F7	Calibration cap and PTFE disc	White reference at the item distance	Disc dirty or worn	Drift in every result	Handling, dust	3	3	2	18	Clean and replace the disc; drift warning at 5 %
F8	Shroud (TPU)	Block direct light and set the distance	Rim worn or torn	Sunlight leak; wrong distance	Abrasion on rough items	3	3	2	18	Replaceable wear part; dark-level prompt catches leaks
F9	Display and sun hood	Show the result	Unreadable in sun	Misread result	Sun near the screen normal	3	3	2	18	Colour and icon cues; hold angle guidance
F10	Slide switch in the battery lead	Turn the scanner on	Cannot charge with the switch off	Flat cell at shift start	Wiring choice	2	3	2	12	Confirm the board's charging path

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
		and allow charging								(WSC-DEC-001 confirm item 3)